# Do Now · retrieval

## Fetch–decode–execute and buses

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

Which register holds the address accessed during a fetch?

### Q2

What does the CIR hold?

### Q3

What kind of information travels along the address bus?

### Q4

A rising-edge D flip-flop has Q=0. D changes to 1 between clock edges; the next rising edge occurs while D=1. State Q before and after that edge, and explain why.

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Fetch uses the buses — The address bus carries the address selected by MAR; memory returns instruction data on the data bus; control signals coordinate the transfer.

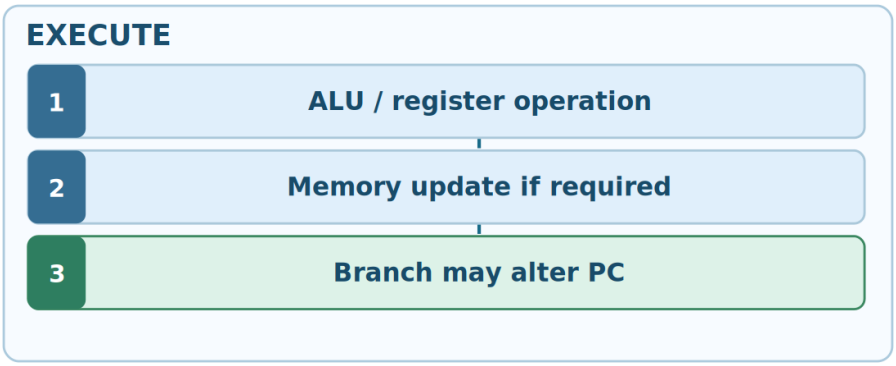
## Core idea 2

Decode interprets instruction — The control unit examines CIR to decide which operation and operands are involved.

## Core idea 3

Execute changes state — An instruction may update ACC or memory, or alter PC for a jump. Interrupt handling is coordinated at appropriate points and can temporarily redirect execution.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.

# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

fetch — Retrieve next instruction from memory.

decode — Interpret opcode and operands.

execute — Perform the operation.

address bus — Transfers memory/device addresses.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

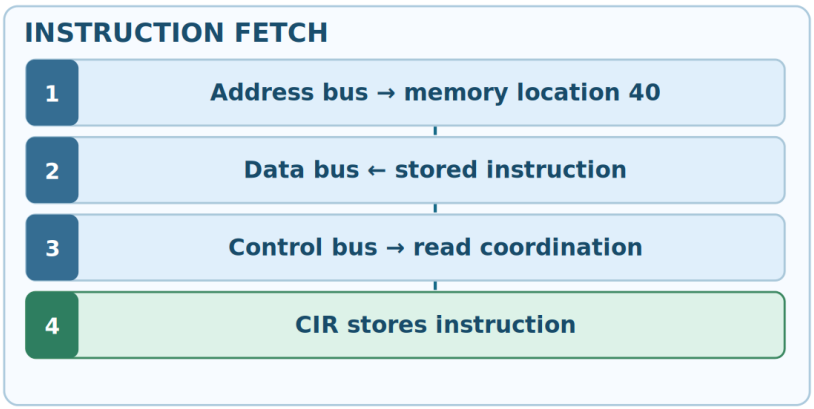
Fetch the instruction at address 40 and explain what each bus contributes.

## Follow the worked method

1. Predict first — MAR identifies location 40; the address bus carries that location to memory, accompanied by a read control signal.

2. Explain the answer — The data bus transfers instruction content to MDR, then CIR; sequential PC becomes 41.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.

# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Predict first

The initial fetch advances PC from 81 to 82, as in a sequential instruction.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

For each bus—address, data and control—state its main function and give one example of the information transported.

## Question 2

Trace a fetch starting PC=64, then explain what happens when CIR contains JMP 100.

Check that you have used the technical terms and explained your decisions.

# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Compare what happens in the instruction cycle during a normal arithmetic instruction and while responding to an interrupt.

Check that you have used the technical terms and explained your decisions.

# Exit Ticket · reflect and retrieve

## Exit Ticket

An unconditional branch can change the Program Counter away from its sequential value.

## Use key vocabulary from memory

1. Explain fetch in your own words.

2. Explain decode in your own words.

3. Explain execute in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.